Dataset Preparation

Machine Learning Pattern Classification

Team Quantized Encoders

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Make the data trustworthy

The task is multi-label, sparse, and based on multiple annotators.
Preparation decides whether later scores measure sound-event recognition or hidden leakage.

Recordings validated	3656	
Target classes	15	
Feature window	1.0s	CNN mel patch: 0.3s before `t_start` + 0.7s after

Label aggregation keeps uncertainty

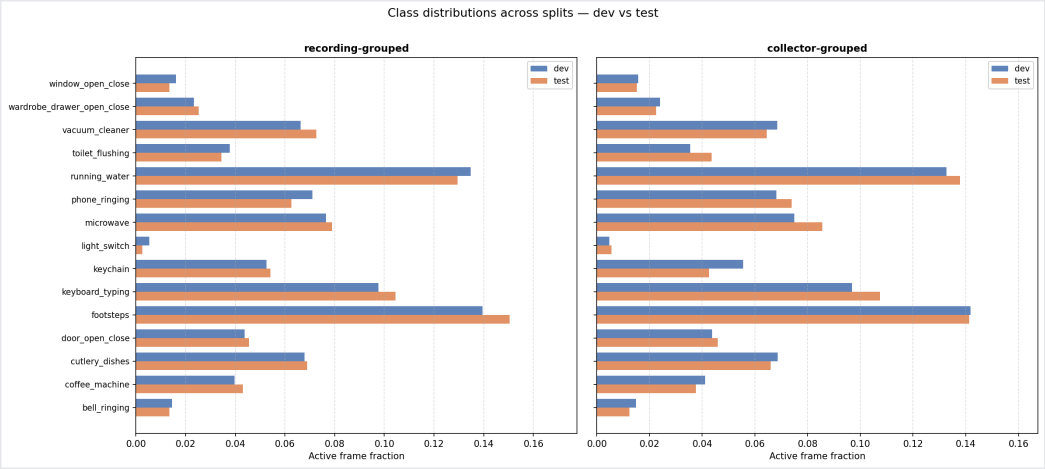
Raw annotation events are projected onto the same frame grid as the features.

- Store labels as `[T, C, A]`
- `T` : feature frames
- `C` : target classes
- `A` : annotators
- Use **max overlap** when one annotator marks repeated same-class events in the same frame

```
soft_labels = annotations.mean(axis=2)  # [T, C]
hard_labels = (soft_labels >= 0.5).astype(int)
```

Soft labels preserve annotator disagreement. Hard labels are only derived when the model needs binary targets.

Splits avoid leakage



Left: recording-grouped. All segments from the same recording stay in one split.

Right: collector-grouped. All recordings from the same collector stay in one split.

Distribution check: dev and test class frequencies should remain comparable.

Final rule: development CV for selection, held-out test only once.

Preprocessing is fit on training data

Conservative baseline pipeline:

1. Convert invalid values to missing values
2. Mean-impute missing feature values
3. **Standardize** all 960 feature dimensions

```
x_scaled = (x - train_mean) / train_std
```

The imputer and scaler are fit only on the training part of each split, then applied to validation or test data.